WEEK 10 – HOUSING DATA DASHBOARD

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INTRODUCTION:

The Week 10 Power BI project was developed as a continuation of the Week 9 project. The cleaned Housing Data dataset was used to perform further analysis and create an interactive Real Estate Analytics Dashboard. The main objective of this project was to analyze house prices, property details, locations, living areas, and other important features.

The dashboard was created using different DAX measures, Card visuals, charts, a map, and a slicer. The report was divided into three pages to present the analysis clearly without overcrowding the dashboard.

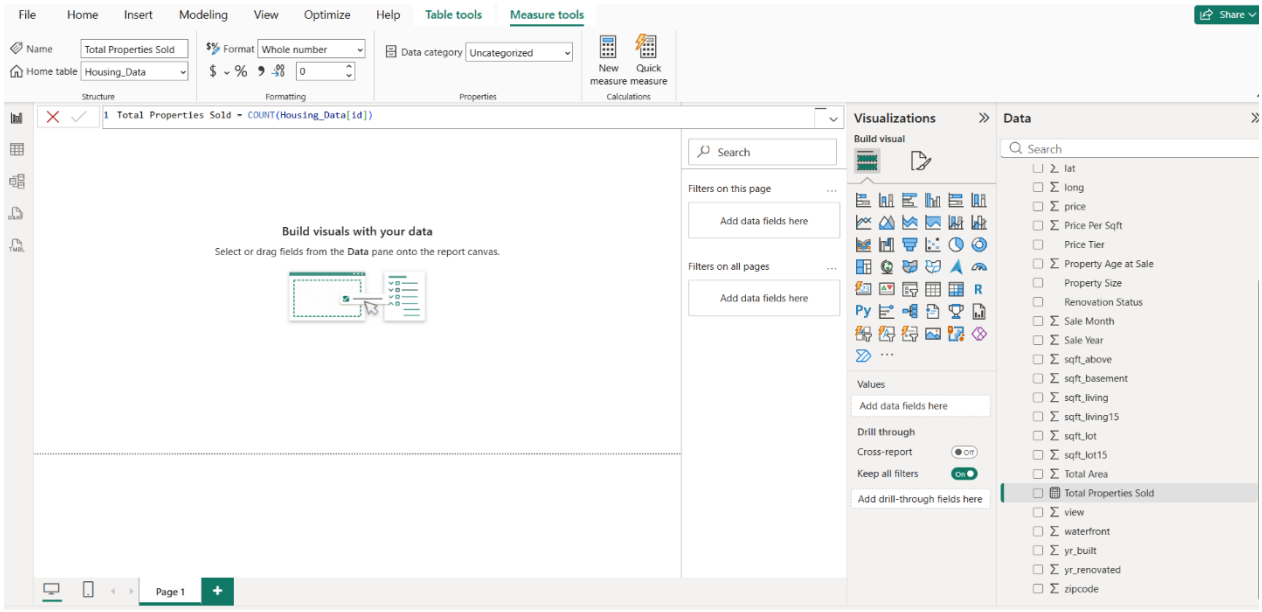
CREATING DAX MEASURES:

The first step in creating the dashboard was to create DAX measures. These measures were used to calculate important information from the Housing_Data table.

The following measures were created:

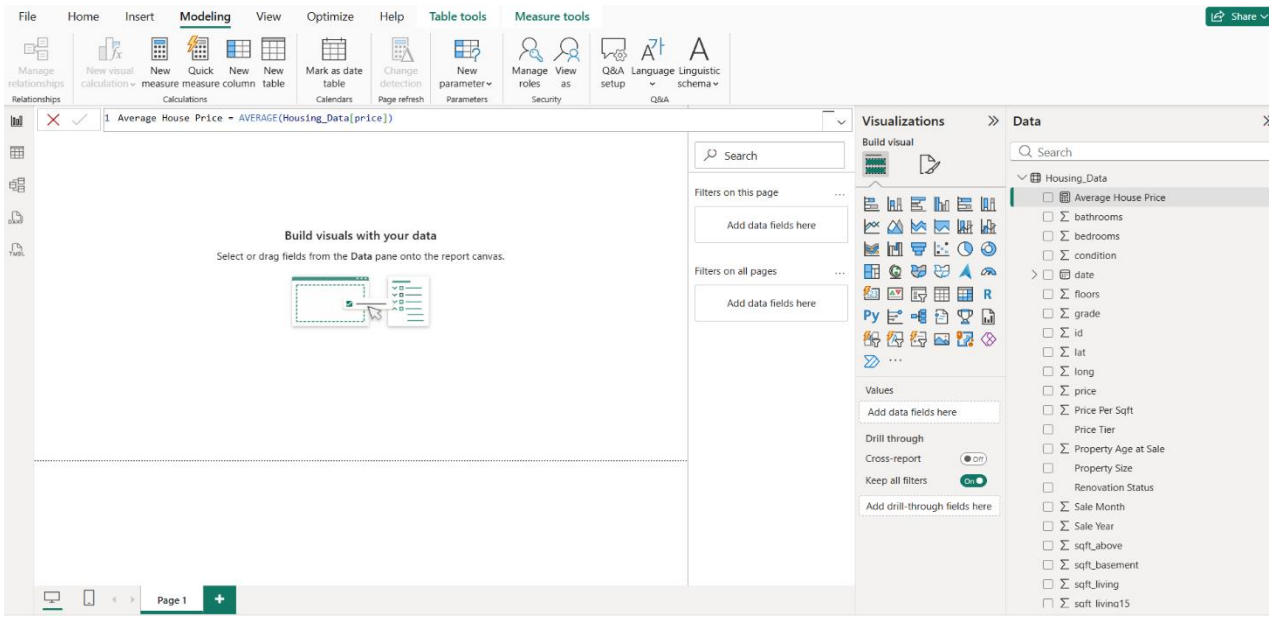
Total Properties Sold

The Total Properties Sold measure was created using the id column to count the total number of properties in the dataset.



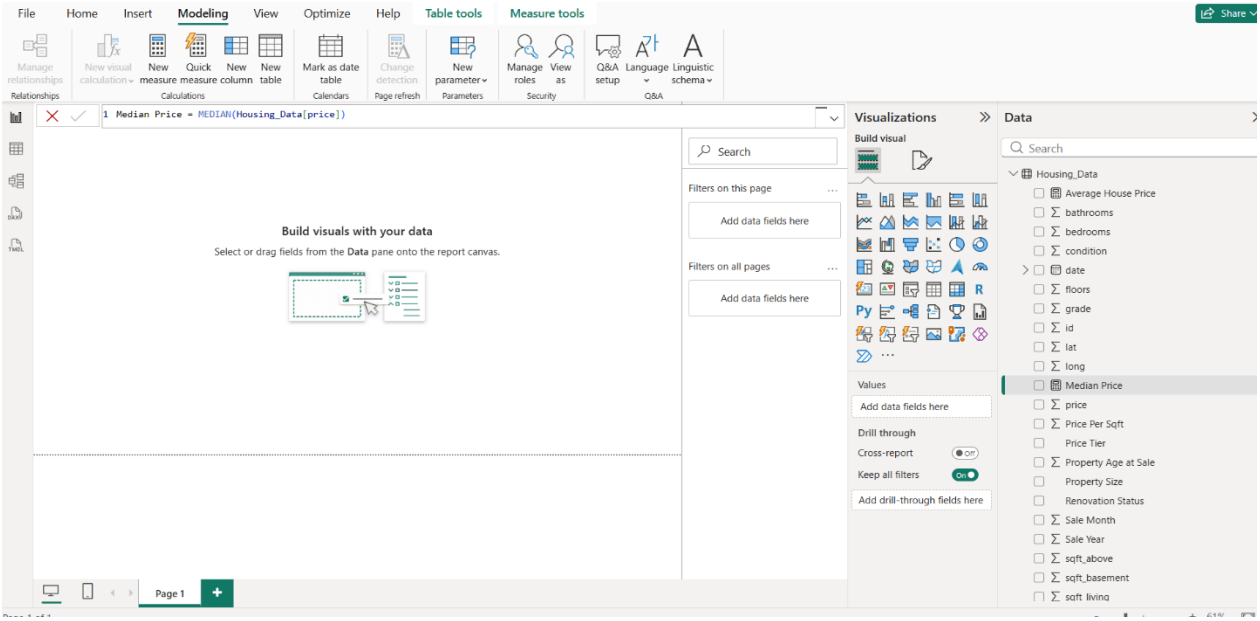
Average House Price

The Average House Price measure was created to calculate the average price of all houses in the dataset.



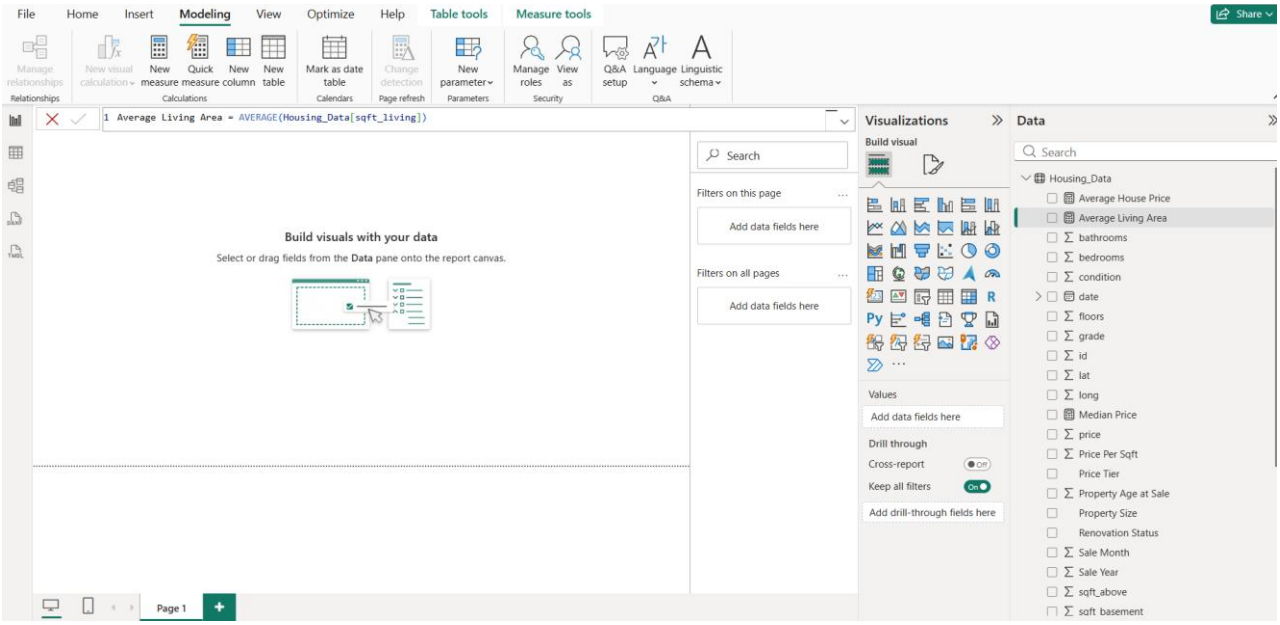
Median Price

The Median Price measure was created to identify the middle value of the house prices.



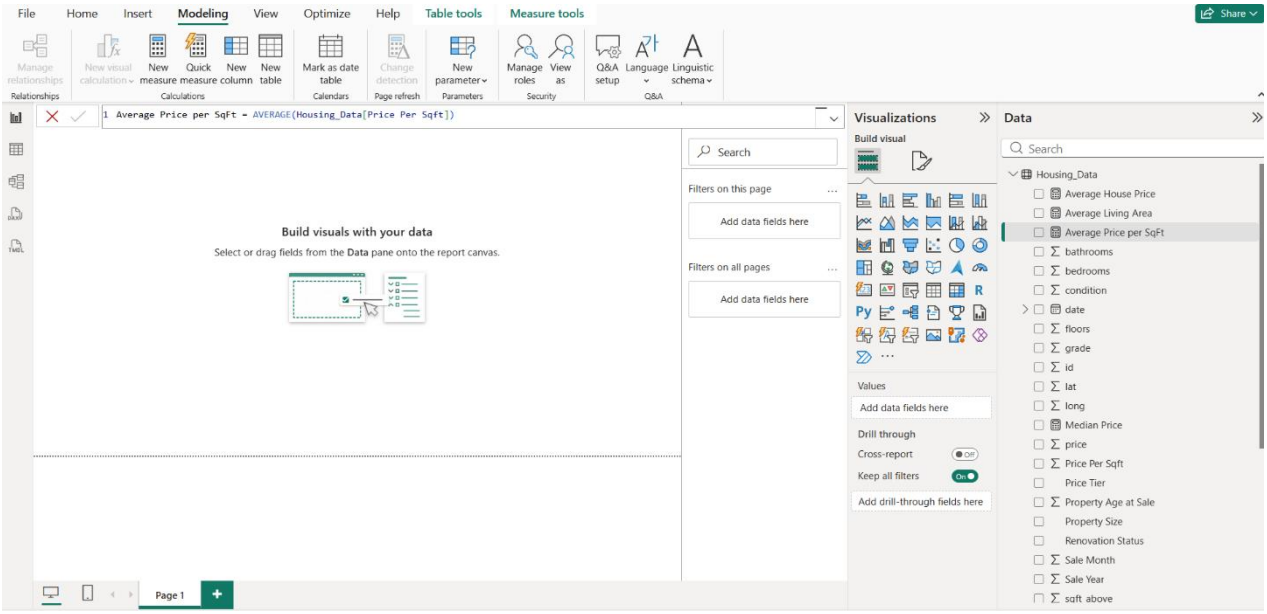
Average Living Area

The Average Living Area measure was created using the sqft_living column. It calculates the average living area of the properties in square feet.



Average Price per SqFt

The Average Price per SqFt measure was created using the existing Price Per Sqft column. This measure calculates the average price per square foot of the properties.

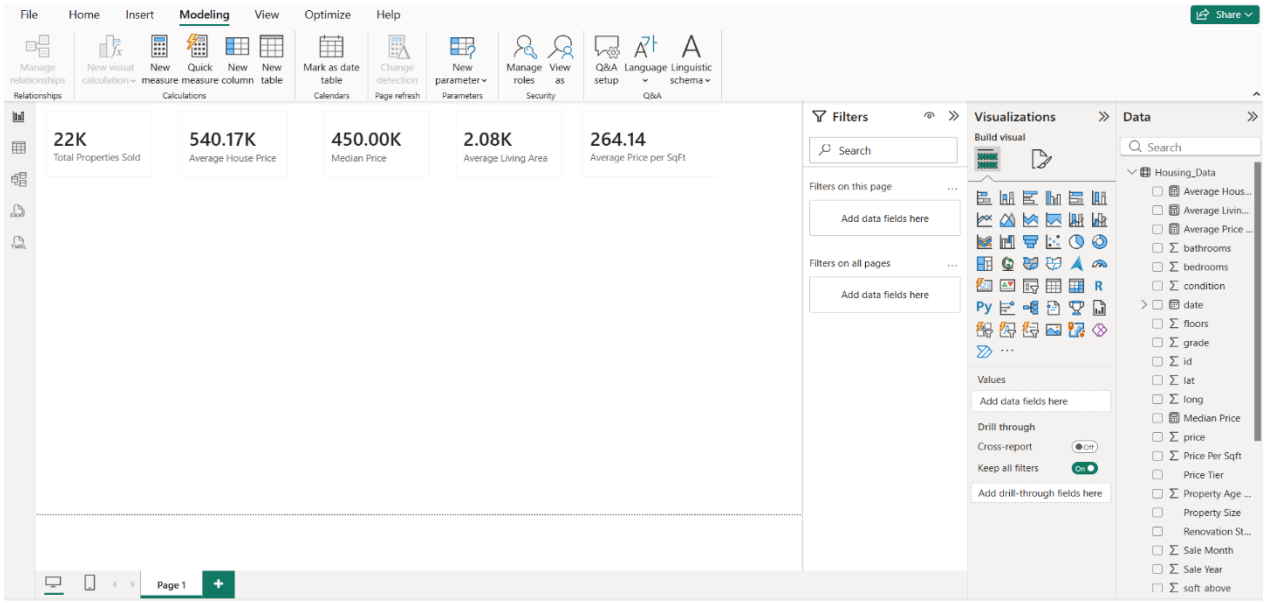


CREATING CARD VISUALS:

After creating the DAX measures, Card visuals were added to display the important real estate values clearly.

Five Card visuals were created for:

- Total Properties Sold
- Average House Price
- Median Price
- Average Living Area
- Average Price per SqFt

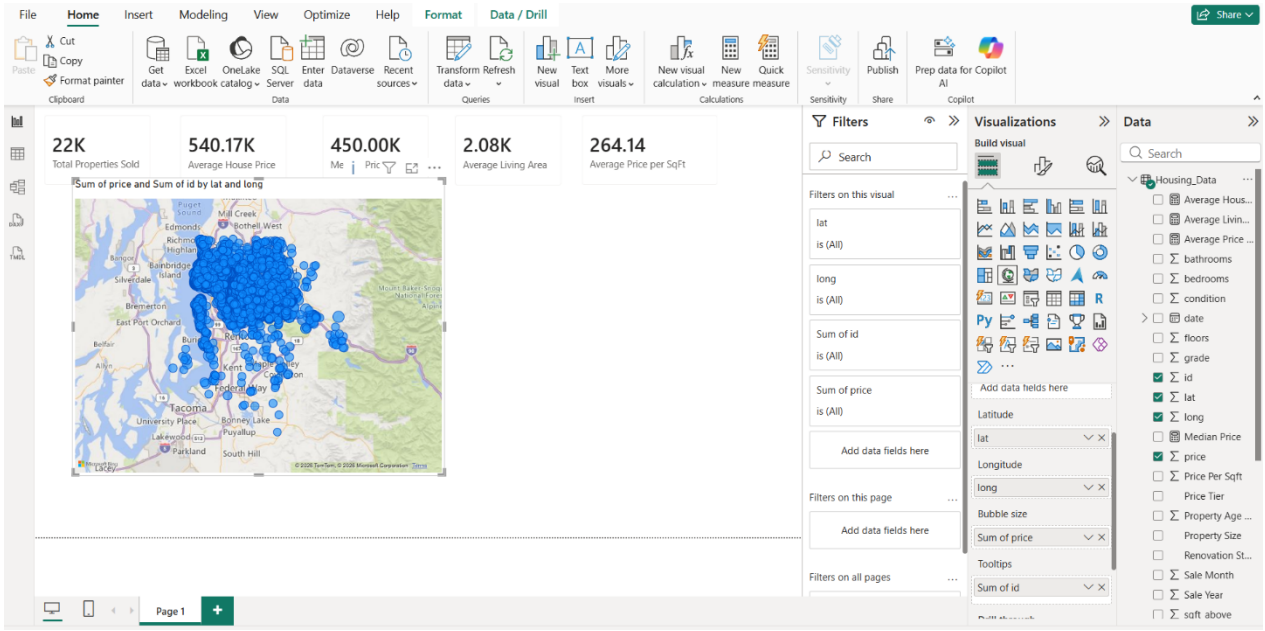


CREATING PROPERTY LOCATION MAP:

A Map visual was created to display the geographical locations of the properties.

The lat column was added to the Latitude field and the long column was added to the Longitude field. The price column was added to Bubble Size so that the size of the bubbles represents the property prices. The id column was added to the Tooltips field.

This visualization helps to understand the geographical distribution and location of the properties.



CREATING COLUMN CHARTS:

CREATING AVERAGE HOUSE PRICE BY BEDROOMS:

A Clustered Column Chart was created to analyze the relationship between the number of bedrooms and the average house price.

The bedrooms column was added to the X-axis and the price column was added to the Y-axis. The price aggregation was changed to Average.

CREATING AVERAGE HOUSE PRICE BY BATHROOMS:

Another Clustered Column Chart was created to analyze the average house price based on the number of bathrooms.

The bathrooms column was added to the X-axis and the price column was added to the Y-axis. The price aggregation was set to Average.

CREATING AVERAGE HOUSE PRICE BY WATERFRONT STATUS:

A Clustered Column Chart was created to compare the average prices of waterfront and non-waterfront properties.

The waterfront column was added to the X-axis and the price column was added to the Y-axis. The price aggregation was changed to Average.

CREATING NUMBER OF PROPERTIES BY GRADE:

A Clustered Column Chart was created to analyze the number of properties according to their grade.

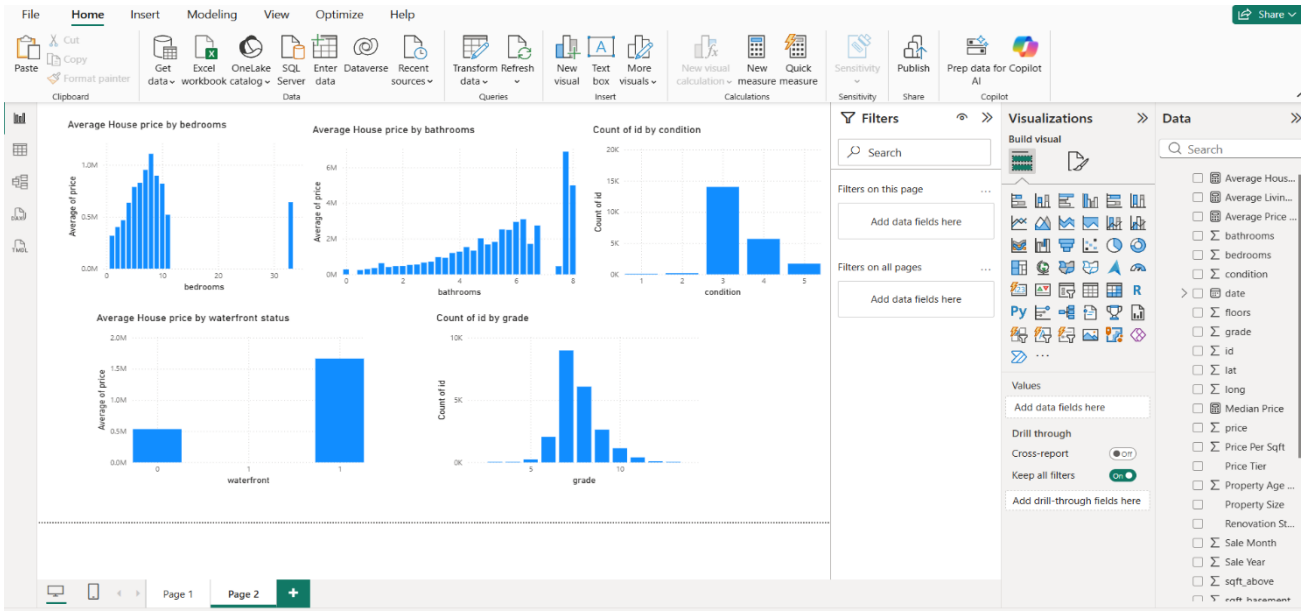
The grade column was added to the X-axis. The id column was added to the Y-axis and its aggregation was set to Count.

This visualization shows how the properties are distributed across different grade categories.

CREATING NUMBER OF PROPERTIES BY CONDITION:

A Clustered Column Chart was created to analyze the number of properties based on their condition.

The condition column was added to the X-axis. The id column was added to the Y-axis and was set to Count.

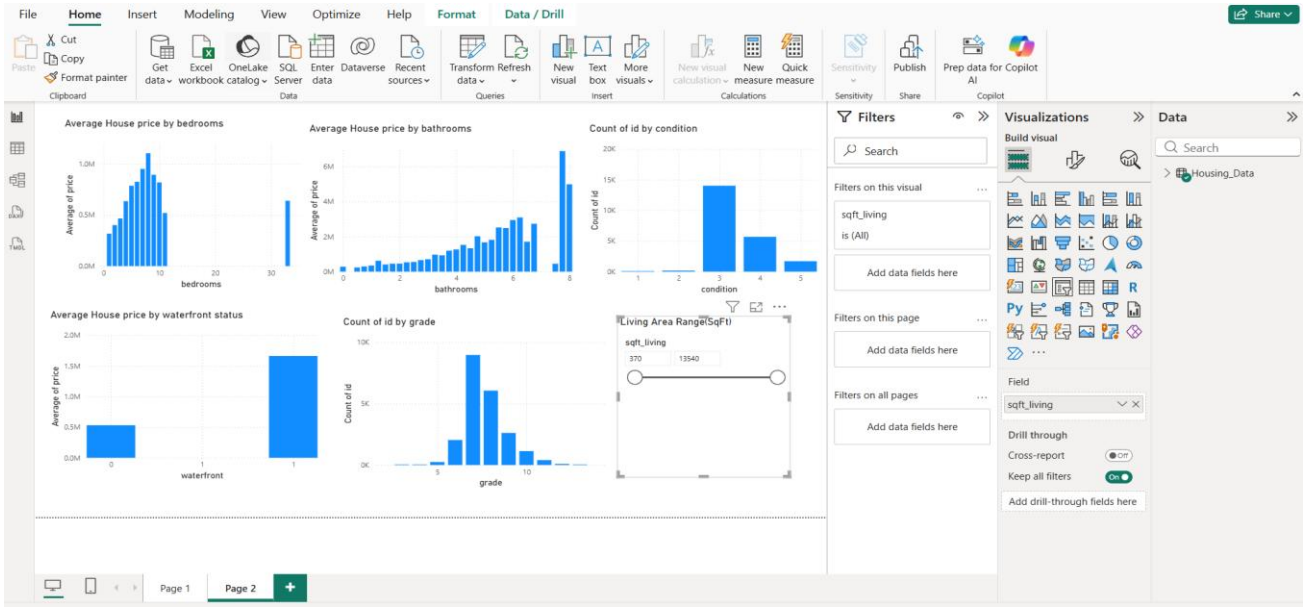


CREATING SLICER:

A Slicer was added to make the dashboard interactive.

The sqft_living column was added to the Slicer visual. Power BI displayed the minimum and maximum living area values as a range slider.

The slicer was titled Living Area Range (SqFt). Users can use this slider to select a specific living area range and filter the property data.



CONCLUSION:

The Real Estate Analytics Dashboard was successfully created using Power BI Desktop.

DAX measures were created to calculate important values such as total properties sold, average house price, median price, average living area, and average price per square foot. Different visualizations were then created to analyze property locations, prices based on bedrooms and bathrooms, waterfront status, property grades, and property conditions.